CRISTIANO ULONDU MENDES

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EDUCATION

PhD in Signal and Image Processing *Palaiseau, France*
Thesis: Deep learning for Synthetic Aperture Radar (SAR) data processing and analysis *2023 -- present*
Topics: speckle filtering, polarimetric SAR despeckling, tomographic SAR reconstruction,
spatially-correlated noise, unrolled algorithms, diffusion models
LTCI, Télécom Paris, Institut Polytechnique de Paris

Master's Degree in Mathematics, Data, and Learning *Paris, France*
Track: Mathematics, Modeling, Learning *2021 - 2022*
Paris Cité University
with highest honour

Bachelor's Degree in Fundamental and Applied Mathematics *Paris, France*
2020 - 2021
Paris Cité University
with high honours

SKILLS

Programming languages:	Python, MATLAB, LaTeX
Frameworks & Libraries:	PyTorch, NumPy, SciPy, OpenCV
Research areas:	Deep learning, inverse problems, sparse reconstruction, variational methods, SAR imaging, polarimetry, tomography
Tools:	Git, GitLab, Linux, HPC/cluster computing
Operating systems:	Linux, Windows

PROJECT

Master 1 semester project: exploration of the state of the art in inverse problem-solving methods with the aim of application in medical imaging.

Demonstrated skills:

- adaptability
- technical rigor
- autonomy and involvement

EXPERIENCE

- End of studies internship *01/03/22 - 31/08/22*
  Laboratoire Traitement et Communication de l'Information (LTCI), Télécom Paris, France

  internship subject: Exploration of variational approaches for multi-temporal despeckling of polarimetric Synthetic Aperture Radar (SAR) images.

  Tasks performed :

  - bibliographic study on SAR imaging

- porting the Matlab code of a single-date SAR image denoiser (MuLoG) to Python
- proposal for an extension to polarimetric SAR images of a multi-temporal denoising scheme initially developed for single-channel SAR images (RABASAR)
- proposal of solutions to account for noise spatial correlation in the despeckling of real SAR images.

Developed skills:

- organizational skills
- autonomy
- ability to analyze scientific articles, adapt, and implement the methods described therein.

- Research engineer *15/10/22 – 31/08/23*
  Laboratoire Traitement et Communication de l'Information (LTCI), Télécom Paris, France

  Subject: Extension of the end-of-studies internship

PUBLICATIONS

Journal Articles

- C. Ulondu Mendes, L. Denis, C. Kervazo, F. Tupin, ``Fast tomographic SAR inversion by unrolling a greedy algorithm for sparse reconstruction,'' *submitted to IEEE Transactions on Geoscience and Remote Sensing*, June 2026. [Preprint: hal-05609504]
- C. Ulondu Mendes, E. Dalsasso, L. Denis, F. Tupin, ``POLSAR2POLSAR: A semi-supervised despeckling algorithm for polarimetric SAR images,'' *ISPRS Journal of Photogrammetry and Remote Sensing*, 2025. DOI: 10.1016/j.isprsjprs.2025.01.008
- C. Ulondu Mendes, L. Denis, C. Deledalle, F. Tupin, ``Robustness to spatially-correlated speckle in Plug-and-Play PolSAR despeckling,'' *IEEE Transactions on Geoscience and Remote Sensing*, 2024. DOI: 10.1109/TGRS.2024.3432180

Conferences & Workshops

- C. Ulondu Mendes, L. Denis, C. Kervazo, F. Tupin, ``Leveraging LiDAR datasets to improve SAR tomography: a diffusion model approach,'' *14th IAPR Workshop on Pattern Recognition in Remote Sensing*, Lyon, France, Aug 2026.
- C. Ulondu Mendes, L. Denis, C. Kervazo, F. Tupin, ``Reconstruction 3D en tomographie radar : apprentissage profond basé sur un Matching Pursuit déroulé,'' *GRETSI 2025*, Strasbourg, France, Aug 2025.
- C. Ulondu Mendes, L. Denis, F. Tupin, ``Schéma Plug-and-Play robuste à la corrélation spatiale du speckle pour le filtrage des données SAR polarimétriques,'' *GRETSI 2023*, Grenoble, France, Aug 2023.